

Option A – Alternating Currents

6. (a) The sinusoidal emf produced by a rotating coil in a magnetic field has a **peak** emf of 18.0 V and a frequency of 70.0 Hz. The coil is connected to the y-input of an oscilloscope.

- (i) The output of the rotating coil is used to power a small lamp of known resistance. Explain how you would calculate the rms input power to the lamp from the known resistance and **peak** emf. [2]

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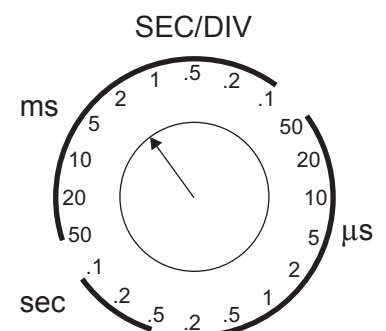
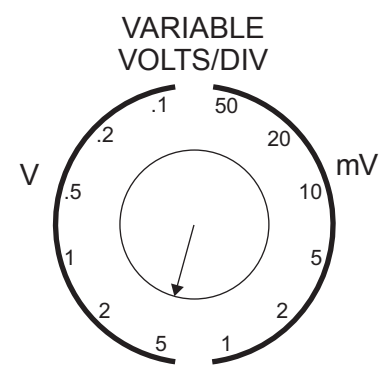
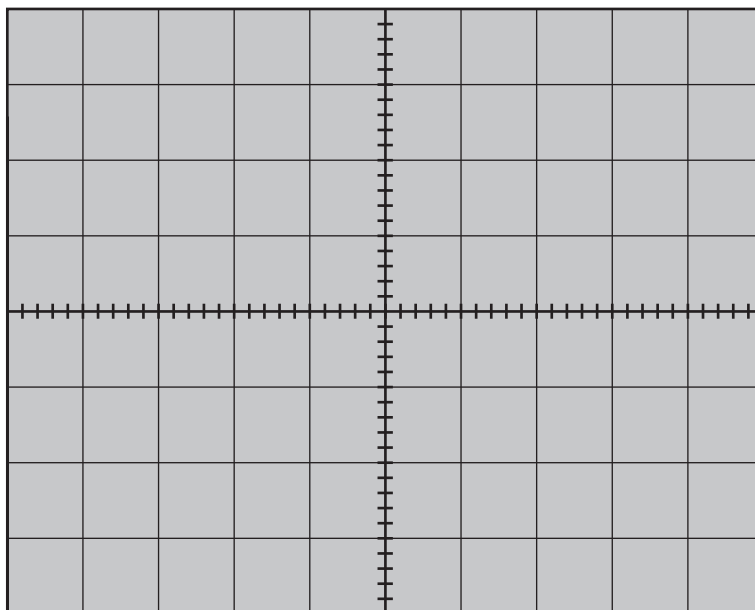
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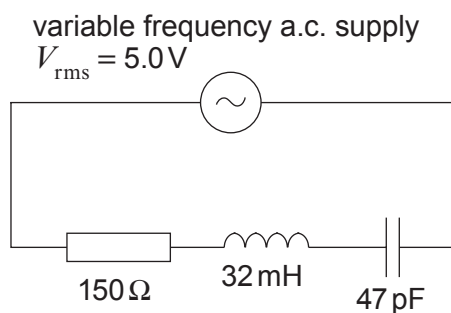
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- (ii) Draw a typical trace that might be seen on the oscilloscope screen with the oscilloscope settings shown. [5]

Space for calculations.



(b) An *LCR* circuit is shown below.



(i) Explain why the resonance frequency of the circuit occurs when:

[3]

$$\omega L = \frac{1}{\omega C}$$

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(ii) Calculate the resonance frequency of the circuit.

[2]

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(iii) Calculate the rms current when the frequency of the supply is 324 kHz.

[3]

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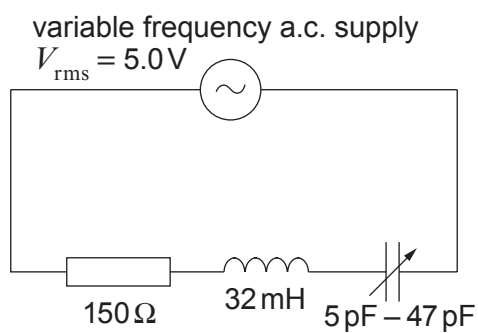
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- (iv) A student claims that the following circuit cannot have a peak pd above 1.5 kV across the capacitor. Investigate whether or not he is correct. [5]

(It may be useful to note that $\frac{\omega_0 L}{R} = \frac{1}{R} \sqrt{\frac{L}{C}}$.)



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